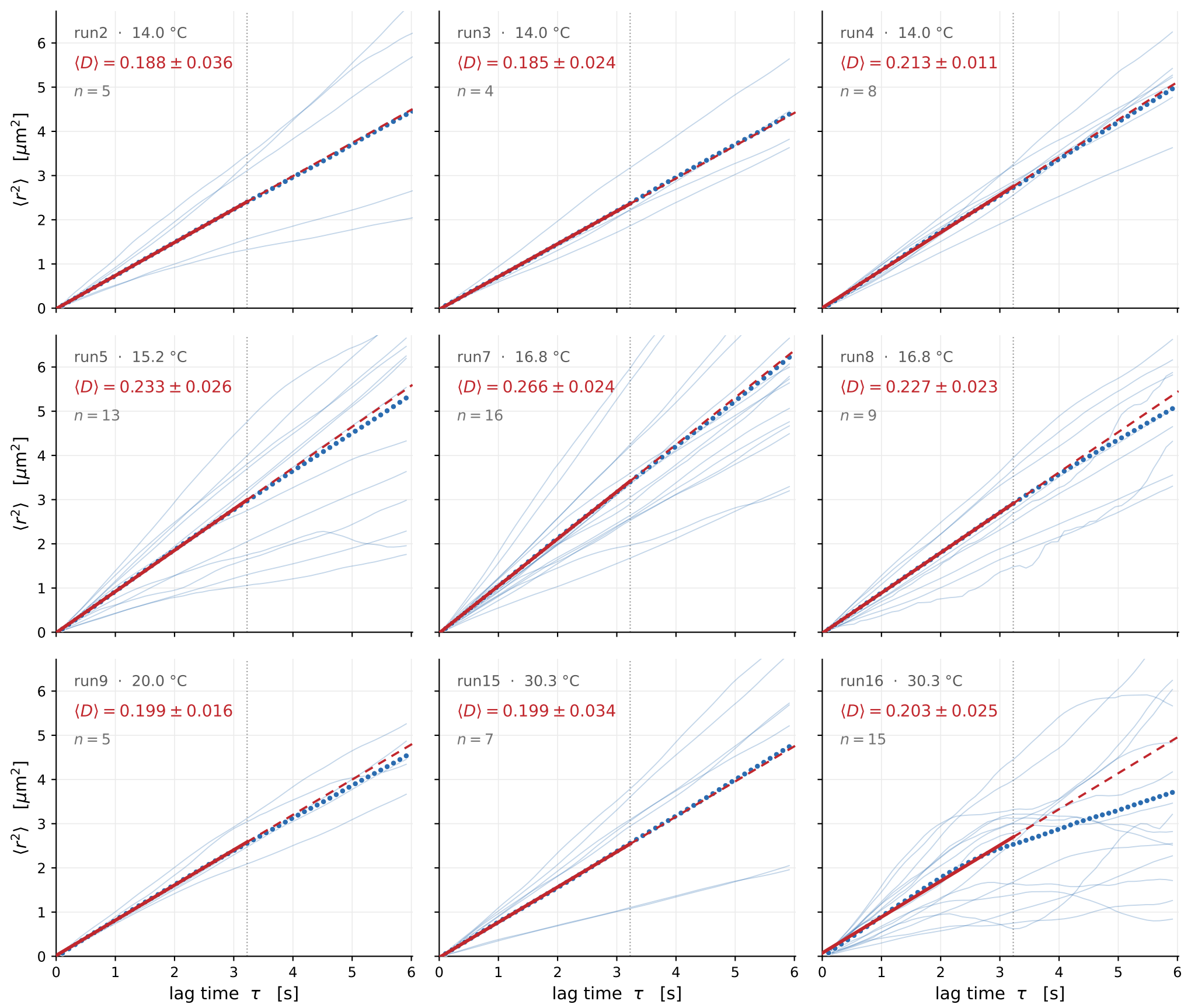




— particle MSD ● ensemble MSD — fit $\langle r^2 \rangle = 4D\tau + c$ - - - extrapolation (excl. from fit)

$\langle D \rangle \pm \text{SE}$ = ensemble-MSD fit over the SHORT-LAG window $\tau \leq 3.2$ s (dotted line; free intercept c , found ≈ 0 — localization $\lesssim 1$ px, so D is unbiased). Fig 2 k_B uses the per-bead median D , which agrees with $\langle D \rangle$ to within 24%.